

all hosts from 192.168.100.1 to 192.168.100.254 by disabling host discovery. (The `-v` option will increase the verbosity of the output.)

A TCP Connect scan

As discussed in the earlier article, any Nmap scan type requiring modification/tweaking of the standard TCP 3-way handshake requires administrative privileges. Since the default Nmap scan is a TCP SYN scan that requires administrator access, for non-privileged users, Nmap uses a TCP Connect scan as the default scan instead. Nmap starts a TCP connect scan by initiating a TCP handshake with a standard SYN packet to the required TCP port of the target device. The target's response to the TCP Connect scan is the same as that in the case of a TCP SYN Scan—it varies, depending on the state of the destination port (see Table 1.1).

Table 1.1: TCP Connect Scan client responses

Port status	Client response	Inference
Open	Standard response: SYN ACK	Service running on the port / port is open
Closed	Standard response: RST	Service not running on the port / port is closed
Filtered	No response	Firewalled port

For a port that is open, Nmap sends an ACK packet to complete the TCP handshake, thus opening the connection. It then sends an RST packet to reset the connection, closing the open connection. In this way, the host is scanned without modifying the standard TCP handshake. Do you see the downside of this scan type? Since it completes the full handshake, it leaves a log entry on the target device.

Like the TCP SYN scan, the TCP Connect scan also works across all operating systems and other devices that implement TCP, such as PLCs, network printers, Ethernet switches, mobile phones, etc. An example scan would be `nmap -sT 192.168.100.100`. The disadvantage of this scan type is that it uses more resources than the TCP SYN scan, since it opens a full TCP connection, and then resets it. It also leaves a log entry on the target device.

Version scan

The more you study Nmap, the more astonished you will be at its functionality. Nmap uses simple TCP, UDP and ICMP scans in very interesting ways—to detect various devices, operating systems, services running on various ports, and even the versions of these services. The Nmap version scan, `-sV`, is used to find the versions of services running on open ports. A simple command like `nmap -v -sV 192.168.100.100` will find open ports on the host, and find and display versions of all the services running on these ports.



CAUTION:

While using Nmap, be warned that under the stringent rules of the Indian Cyber Law 2000 and its further amendments till date, even a port scan on a public IP could land you in jail. Do not scan computers that you do not own; nor run scans over networks that you do not own, without written permission from the owners.

You may scan scanme.nmap.org for testing. This permission only includes scanning via Nmap; you are not allowed to test exploits and/or denial of service attacks. Don't forget to follow the rules listed in the Nmap man page. Abuse of this service will be reported to the government by the site owners.

Use Nmap very carefully, and only for discovery/audits of your network. As mentioned above, it is a very powerful tool, and could cause disruption/damage to the target system or network. Fully understand what you are doing (even with scripted scans), before you do it. Ignorance will not excuse anybody from being prosecuted under the law.

Also note that if you have a website of your own, either hosted at a hosting provider or on a rented physical server, the server and network do NOT belong to you even though you own the website's content. You should ideally obtain permission from such hosting providers/ server owners to carry out even "testing" probes of your own website..

OS fingerprinting

Nmap can detect the operating system of the scanned host. If, for example, you use `nmap -v -O 192.168.100.100`, it can detect a variety of operating systems like the versions of Microsoft Windows, Linux (including kernel version), etc. If there is an unrecognised service or operating system, Nmap will show the service's fingerprint, and request the user to send the fingerprint and the version of the service/operating system to the Nmap developers at www.insecure.org, for inclusion in future Nmap versions.

Please try out these scanning techniques, hands-on, before further exploring various other scan options provided by Nmap. And don't forget to keep a watch on this column for further details! 

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